

Homological Projective Duality

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Conventions. For a $\mathbb{C}$ vector space E , we regard $\mathbb{P}(E)$ to be the projective space of *lines* in E , so

$$\mathbb{P}(E) = \text{Proj Sym}^\bullet E^*.$$

When V is a basepoint free linear system for some line bundle L on X , this projective space $\mathbb{P}(V^*)$ is the natural codomain for a morphism from X . On $\mathbb{C}$ points, this is just sending

$$x \longmapsto \mathbb{C} \cdot \text{eval}_x,$$

i.e., we send x to the “line” in the dual space which is scalar multiples of the functional on sections of V which is evaluating them at x . This is to be expected—sections of L are coordinates, and we usually want to take Proj of something resembling coordinates, and indeed

$$\mathbb{P}(V^*) = \text{Proj Sym}^\bullet V,$$

(at least when V is finite dimensional) so that $\text{Sym}^\bullet V$ is like coordinates.

1 Setup

Fix X a variety and line bundle $\mathcal{O}_X(1)$ with basepoint free linear system $V \subseteq H^0(\mathcal{O}_X(1))$ giving a map $f : X \rightarrow \mathbb{P}(V^*)$. In $\mathbb{P}(V^*) \times \mathbb{P}(V)$ we get the incidence correspondence $\mathcal{H}$ which has as points x, H where $x \in \mathbb{P}(V^*)$ is a point lying on the hyperplane H in $\mathbb{P}(V^*)$ (so a point in $\mathbb{P}(V)$). We pullback the incidence correspondence to $X \times \mathbb{P}(V)$

$$\mathcal{H} = \{(x, s) \mid s(x) = 0\} \subseteq X \times \mathbb{P}(V)$$

which is cut out by a section of

$$H^0(\mathcal{O}_{X \times \mathbb{P}(V)}(1, 1)) \cong H^0(\mathcal{O}_X(1)) \otimes V^* \cong \text{Hom}(V, H^0(\mathcal{O}_X(1)))$$

which corresponds to the inclusion $V \hookrightarrow H^0(\mathcal{O}_X(1))$.

We should think of $\mathcal{H} \rightarrow X$ as a family, the fiber over $x \in X$ is the set of hyperplanes of $\mathbb{P}(V^*)$ containing x .

Now, we can restrict V to a maybe not basepoint free subsystem $L \leq V$ of dimension ℓ , and set $L^\perp \subseteq V^*$ its annihilator. Then, $\mathbb{P}(L^\perp) \subseteq \mathbb{P}(V^*)$ are the points contained in every hyperplane in L . Then, set

$$\begin{aligned} X_{L^\perp} &:= X \times_{\mathbb{P}(V^*)} \mathbb{P}(L^\perp), \\ \mathcal{H}_L &:= \mathcal{H} \times_{\mathbb{P}(V)} \mathbb{P}(L). \end{aligned}$$

In this way, $X_{L^\perp}$ is the points of X contained in every hyperplane in L , i.e., $X_{L^\perp}$ is the baselocus of L . Then, $\mathcal{H}_L$ is a family over X , where the fiber over $x \in X$ is the set of hyperplanes in L containing $x \in X$.

As we consider the family $\mathcal{H}_L \rightarrow X$, we get the diagram

$$\begin{array}{ccccc} X_{L^\perp} \times \mathbb{P}(L) & \xhookrightarrow{j} & \mathcal{H}_L & \xhookrightarrow{\iota} & X \times \mathbb{P}(L) \\ \downarrow p & & \downarrow \pi & \swarrow \rho & \\ X_{L^\perp} & \xhookrightarrow{i} & X & & \end{array}$$

In this diagram, we observe that generally $x \in X$ has fiber which is $\ell - 2$ dimensional (i.e., in the $\ell - 1$ dimensional space $\mathbb{P}(L)$ of hyperplanes, vanishing at x is one equation which is generally nontrivial). However, along $X_{L^\perp}$, all hyperplanes in $\mathbb{P}(L)$ contain x , so we get the full $\ell - 1$ dimensional fiber on this closed subset.

Proposition 1.1. Suppose $X_{L^\perp}$ has dimension $\dim X - \ell$ (so it is a complete intersection). Then,

$$j_* p^* : D(X_{L^\perp}) \rightarrow D(\mathcal{H}_L)$$

is a fully faithful embedding, and we get a semiorthogonal decomposition

$$D(\mathcal{H}_L) = \langle D(X_{L^\perp}), \pi^* D(X)(0, 1), \dots, \pi^* D(X)(0, \ell - 1) \rangle$$

(for twisting, the first argument is from $\mathcal{O}_X(1)$, the second is from $\mathcal{O}_{\mathbb{P}(L)}(1)$).

One should also check that π^* is fully faithful because unit $1 \Rightarrow R\pi_* \pi^*$ is an isomorphism, where checking this boils down to checking $R\pi_* \mathcal{O}_{\mathcal{H}_L} \cong \mathcal{O}_X$ and then tensoring/using projection formula (maybe need smoothness assumptions?).

Part of the point here is that the usual Beilinson type exceptional collection for $\mathbb{P}(L)$ would involve twists $\mathcal{O}, \dots, \mathcal{O}(\ell - 1)$, but here we did not put $\mathcal{O}$, which was instead covered by $D(X_{L^\perp})$. The point then is that as we bleed into $D(X_{L^\perp})$ by writing down the “zeroth twist”, we get some interesting overlap.

Assume that $D(X)$ has a *rectangular Lefschetz decomposition*, so there is an admissible subcategory $\mathcal{A} \subseteq D(X)$ with

$$D(X) = \langle \mathcal{A}, \mathcal{A}(1), \dots, \mathcal{A}(\ell - 1) \rangle$$

which lets us refine the proposition to get a decomposition

$$D(\mathcal{H}_L) = \langle D(X_{L^\perp}), \mathcal{A}(0, 1), \dots, \mathcal{A}(0, \ell - 1), \mathcal{A}(1, 1), \dots, \mathcal{A}(1, \ell - 1), \dots, \mathcal{A}(\ell - 1, \ell - 1) \rangle.$$

Now, we screw with this and take some twists of the form $\mathcal{A}(j, 0)$, so we take a “zeroth twist” for $\mathbb{P}(L)$, but we’ll also make $j \geq 1$ to ensure semiorthogonality (i.e., we don’t want to write down too many pieces so that they aren’t orthogonal anymore). We can abbreviate notation by recognizing

$$\langle \mathcal{A}(j, 0), \dots, \mathcal{A}(j, \ell - 1) \rangle = \mathcal{A} \boxtimes D(\mathbb{P}(L)),$$

and then we can prove that

$$\langle \mathcal{A}(1) \boxtimes D(\mathbb{P}(L)), \dots, \mathcal{A}(i - 1) \boxtimes D(\mathbb{P}(L)) \rangle$$

is a semiorthogonal collection, but some is left over. Define the right orthogonal complement to be $\mathcal{C}_{\mathcal{H}_L}$.

Homological projective duality then tells us how $\mathcal{C}_{\mathcal{H}_L}$ and $D(X_{L^\perp})$ are built out of each other as the dimension of L grows and shrinks.